LDGAN: A Synthetic Aperture Radar Image Generation Method for Automatic Target Recognition

Changjie Cao, Zongjie Cao, *Member, IEEE*, and Zongyong Cui, *Member, IEEE*

***Abstract*— Under the framework of a supervised learning-based automatic target recognition (ATR) approach, recognition performance is primarily dependent on the amount of training samples. However, shortage in training samples is a consistent issue for ATR. In this article, we propose a new image to image generation method, called label-directed generative adversarial networks (LDGANs), which will provide labeled samples to be used for recognition model training. We define an entirely new loss function for the LDGAN, which utilizes the Wasserstein distance to replace the original distance measurement of the conventional generative adversarial networks (GANs), thus efficiently avoiding the collapse mode problem. The label information is also added to the loss function of the LDGAN to avoid generating a large number of unlabeled target images. More importantly, the proposed method also makes corresponding changes to the network architecture regarding the new GANs. At the same time, the detailed algorithm about the LDGAN is also introduced in this article to deal with the issue that characteristically GANs are not easy to train. Based on comparisons with other directed generation methods, the experimental results show comparative results of several types of generated images in statistical features, gradient features, classic features of synthetic aperture radar (SAR) targets and the independence from the real image. While demonstrating that the images generated by the LDGAN produced better results using the assumptions of independent and identical distribution, the experiment also explores the performance of the generated image in the ATR. A comparison of these experimental results demonstrates a better way to use the generated image for ATR. The experimental results also prove that the proposed method does have the ability to supplement information for ATR when the training sample information is insufficient.**

***Index Terms*— Automatic target recognition (ATR), information supplement, label-directed generative adversarial network (LDGAN), synthetic aperture radar (SAR) image.**

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I. Introduction

THERE are many potential applications for the synthetic aperture radar (SAR) images recognition in military and homeland security. It has been proven that implementing data-driven recognition algorithms for the automatic target recognition (ATR) of the SAR image is effective [1]–[4].

The use of data-driven SAR image recognition algorithms has a long history which can be traced back through the literature regarding its development. When supervised learning for the SAR recognition is mentioned, a variety of literature contains vast documentation and referencing regarding data-driven recognition algorithm. For example, based on the training samples, when a support vector machine (SVM) is obtained by selecting support vectors, we can construct the training of the SVM model to implement SAR recognition [5], [6]. It is generally more common to use feature extraction algorithms such as principal component analysis (PCA) [7]–[9], nonnegative matrix factorization (NMF) [10], and linear discriminant analysis (LDA) [11], [12] to solve the SAR image recognition. With more and more attention being paid to convolutional neural networks (CNNs) in recent years, the research focus of the SAR image recognition has gradually shifted from conventional statistical machine learning algorithm (such as PCA, NMF, LDA, and SVM) to deep learning. Chen *et al.* [13] utilized deep convolutional networks for ATR of SAR images, and as a result, were able to get high accuracy of the SAR image recognition. Automatic recognition technology combined by CNN and SVM is also applied to SAR image recognition in [14]. Zhou *et al.* [15] extended CNN to the field of polarimetric SAR image recognition.

The data-driven recognition algorithms rely heavily on the capacity of the data set [16]. In particular, for the deep neural network algorithm, sufficient sample size is foundational to its feasibility. However, the shortage of labeled data sets with real SAR images of a high enough capacity has become a common problem. Thus, the lack of the SAR image data sets gradually limits the development of deep learning in the SAR image recognition fields. In the area of the SAR image recognition, the limited sample data set capacity has become increasingly exposed. Thus, data synthesis and data augmentation play a big role in the development of ATR of SAR to overcome this problem. Malmgren-Hansen *et al.* [17] utilized generated data information to improve the ATR of SAR with transfer learning. This article demonstrated that a CNN pretrained on generated

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data had a more significant advantage over a CNN trained only on real image data. Ding *et al.* [18] adopted a multilevel reconstruction using attributed scattering centers to implement data augmentation for SAR target recognition.

It is natural for SAR image recognition to develop image simulators based on computer-aided drawing (CAD) models. However, the accuracy of CAD models is a result of geometric error and simplification in the electromagnetic calculation [13]. More importantly, the quality of the SAR image directly affects the recognition accuracy. Thus, Guo *et al.* [19] proposed a simulation method based on generative adversarial networks (GANs), which do not require any prior assumptions for the target. At the same time, this article also showed the massive potential of GANs in SAR applications.

With the rapid development of GANs, many applications of GANs in the SAR images have emerged in the past 2 years. Merkle *et al.* [20] explored the potential of conditional GAN for optical and SAR image matching. Gao *et al.* [21] utilized the deep convolutional generative adversarial networks (DCGANs) to implement SAR image recognition. Schwegmann *et al.* [22] applied information maximizing generative adversarial networks (InfoGANs) to SAR ship images discrimination and generation.

Since Guo *et al.* [19] proposed to implement SAR image generation using GANs, only a few articles have focused on SAR image generation, even though the quality of SAR image generation directly affects SAR image application. Therefore, this leaves ample room for the development of GANs to generate the SAR images. Guo *et al.* stimulated the possibility of GANs generating SAR images, but the method still needs to improve for two reasons. First, collapse mode may occur in the process of using GAN to generate the SAR images. Collapse mode is defined as the mode where GANs only generate a single type of sample, and the generated samples lack needed diversity. Second, the use of generated images for the SAR image application has not been verified, which is the initial purpose of image generation.

The practice of generating a natural image via GANs model is currently a research hotspot in deep learning. When using GANs to generate images, collapse mode may occur in the process. Arjovsky *et al.* [23] proposed using Wasserstein GAN (WGAN) to overcome the problem of collapse mode of GANs. In addition, this article also proves that the conventional GANs loss function is not a reasonable distance measure from mathematical theory. Conversely, the Wasserstein distance makes the GANs avoid the problems of gradient vanish when fitting the generators distribution and the real data distribution. We have adopted the Wasserstein distance to replace the distance measure of the loss function of conventional GANs, demonstrating an effective solution for the problem of collapse mode in our article. However, it is not easy to achieve convergence of the discriminator of WGAN, gradient explode may occur in the process of image generation using WGAN. Then, Gulrajani *et al.* [24] proposed an improved WGAN's method, which applied a gradient penalty (GP) to WGAN. While avoiding the collapse mode problem of GANs, the problem of nonconvergence of the discriminator of WGAN is also effectively solved. Subsequently, Zhang *et al.* [25] utilized the improved WGAN, called WGAN-GP, to implement the SAR image generation. More importantly, this article confirmed the feasibility of the generation of the SAR images through GAN to be used for recognition. At present, articles using GAN to generate the SAR images all indicate that the azimuthal angle out of the known data set may occur in the process of the SAR image generation. There is a real possibility for supplementing information. Therefore, it is reasonable to attempt using GANs to potentially generate the SAR images for data set expansion. However, when Zhang *et al.* used WGAN-GP to generate the SAR images, the images were generated randomly, which only generated a large number of unlabeled SAR images. Moreover, large unlabeled sample sets are not enough for the ATR in supervised learning.

This article proposes an alternative SAR image generated method based on label-directed GANs (LDGANs). This method applies the LDGAN architecture to control the generation of SAR image data of a certain class. Although this does achieve the goal of directionally expanding our data set, now it does not require manual or training models to label the generated image. Moreover, this also provides a potential for the augmentation of original database information. In Section II, related work to GANs and its extension will be presented. In Section III, all the details of LDGANs are proposed and subsequently described. A series of experimental results are used to prove the effectiveness and potential of the LDGAN in Section IV. Finally, Section V presents the conclusions of our study.

II. Related Work

A. WGAN

GANs were first proposed by Goodfellow *et al.* [26]. The architecture of GANs consists of a deconvolutional network and a convolutional network. The deconvolutional network, called generator G, is trained to generate samples from data distribution $P_r(x)$, by changing the noise z from noise distribution $P_z(z)$ to $P_r(x)$, where $x = G(z)$. The convolutional network, called discriminator D, is trained to determine whether the sample generated by the generator is a real or fake sample during the process of training. The G and D are represented by the multilayer perceptron (MLP). The generated samples attempt to deceive D to the greatest extent in order to affect the output of D. The D is used to maximize the probability of assigning the correct label to both training samples and generated samples. All in all, D and G look like playing the two-player minimax game on the condition of the following function $V(D, G)$:

$$\min_G \max_D V(D,G) = E_{x\sim P_r(x)}[\log D(x)] + E_{z\sim P_z(z)}[\log(1 - D(G(z)))] \quad (1)$$

where $D(x)$ is the probability that the sample x comes from the data distribution $P_r(x)$ rather than the generators distribution $P_g(x)$.

With the rapid development of GANs, Radford *et al.* [27] proposed the DCGANs by historical attempts. The core of

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DCGAN adopts and modifies three changes to CNN architectures by GANs. First, DCGAN uses the architecture of all convolutional net to replace the spatial pooling functions of CNN. Next, when the convolution feature extraction of DCGAN is completed, the full connection layer in the convolution network architecture is removed. Finally, the batch normalization is also adopted to solve the problem of collapse mode. The proposal of DCGAN promotes the extensive application of GANs in the field of the image.

In practice, DCGAN can only alleviate the collapse mode problem, but it cannot solve this problem fundamentally. As confirmation of this issue, Ding *et al.* [18] used a series of evidences to prove that GAN is an unreasonable distance measurement. Thus, we have utilized another kind of distance measure, called Wasserstein distance, to replace the original distance measure. The formulation of Wasserstein distance is as follows:

$$W(P_r(x), P_g(y)) = \inf_{\gamma \in \prod(P_r(x), P_g(y))} E_{(x,y)\sim\gamma}[\|x - y\|] \quad (2)$$

where $\prod(P_r(x), P_g(y))$ is the set of all the joint distributions $\gamma(x, y)$, and its marginal distribution are $P_r(x)$ and $P_g(y)$, respectively. Equation (2) cannot be solved directly, thus the loss function of WGAN can be transformed as follows:

$$L \approx \max_{\omega:\|D_\omega\|_L \le K} E_{x\sim P_r(x)}[D_\omega(x)] - E_{x\sim P_g(x)}[D_\omega(G(x))] \quad (3)$$

where the $\|D_\omega\|_L \le K$ indicates that the optimal solution of (3) must be established on the basis of the Lipschitz constant $\|D_\omega\|_L$ of D not more than the constant K. ω represents the parameters of the D.

The WGAN easily addresses the problematic appearance of the gradient vanish or gradient explode in the training process. Gulrajani *et al.* [24] demonstrated that it is not enough that the constraint condition satisfies only the Lipschitz constant $\|D_\omega\|_L$ of not exceeding K. This article enforces the 1-Lipschitz constraint to the WGAN. Thus, the loss function of WGAN with GP is as follows:

$$L = E_{x_g\sim P_g}[D(x_g)] - E_{x_r\sim P_r}[D(x_r)] + \lambda E_{x\sim P_x}[(\|\nabla_x D(x)\|_2 - 1)^2] \quad (4)$$

where x is calculated by $x = \epsilon x_r + (1-\epsilon)x_g$. λ is a constraint constant.

B. Conditional GAN

With enormous strides of success in the field of image generation using GANs, the fact that GANs cannot control the class of images being generated is finally being noticed. Then Mirza and Osindero [28] proposed a conditional GAN to direct the image generation in the training process. In the conditional GAN, the conditional information includes, but is not restricted to just class labels.

The conditional GAN utilizes the condition information, such as class labels, to combine the initializing noise so that it can add additional information for the G and D. The subsequent experiments of this article also demonstrate additional information can direct image generation. At the same time, conditional GAN provides another different way

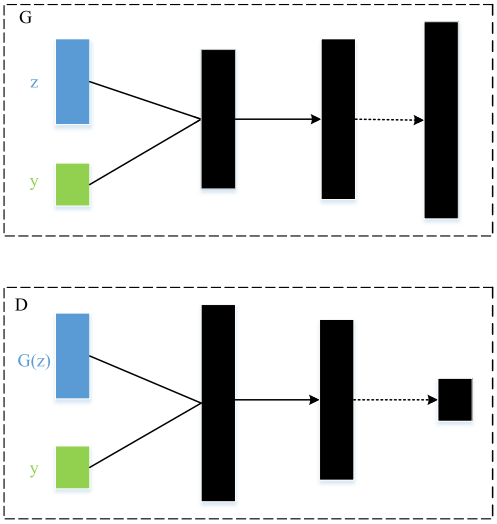


Fig. 1. Schematic of the architecture of conditional GAN. The conditional GAN consists of the generator and the discriminator. In the generator, blue rectangle indicates initialized noise z. The label information y and the MLP are represented by a green rectangle and black rectangles, respectively. In the discriminator, blue rectangle indicates the generated images. Other components are the same as the representation in the generator.

of thinking about image generation. The loss function of conditional GAN is as follows:

$$\min_G \max_D V(D, G) = E_{x\sim P_r(x)}[\log D(x|y)] + E_{z\sim P_z(z)}[\log(1 - D(G(z|y)))]. \quad (5)$$

More importantly, the conditional GAN also changes the model architecture of the original GAN. The schematic of the architecture of conditional GAN is as shown in Fig. 1. G and D are represented by the MLP. z and $G(z)$ indicate the initialized noise and the generated samples, respectively. The conditional information is represented by y.

Subsequently, Gauthier [29] proposed a convolutional way to implement the conditional GAN. The author chose a particular axis and implemented the constant additive shifts along the selected axis of the conditional information y, which produced excellent experimental results. According to the modification of senior axis, the generated faces are obviously aged.

The above article also provides some inspiration for our current work. At present, there is limited literature about the directional generation of a SAR image through the modification of conditional information. However, the directional generation of images is of great value and practical significance. When GAN is adopted to generate SAR images, it is not enough to make sure that the generated images are available. A large number of unlabeled generated images cannot be used directly for data-driven supervised learning. All generated images also require manual or training machine models for labeling, which is unnecessary and uneconomical. Consequently, this process also faces the risk of making mistakes in labeling the generated images. It is of high value to propose a more efficient and convenient method to replace the current one.

According to the previous work, it is easy to note that the modification or addition of conditional information may affect

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the results of image generation. More importantly, based on the experimental results, the GAN generates real samples that do not exist in the original sample [25]. The experimental results also show the possibility of generating SAR image directionally.

Considering the current dearth of literature that provides research on conditional information to generate images used in the SAR image generation model, our work provides a starting point for the use of directional generation of the SAR image using conditional information. There are many kinds of conditional information of SAR image generation. In this article, our approach pays more attention to utilization of label information to generate SAR images directionally. At the same time, we addressed the generation of SAR images with stronger practicability.

III. LDGAN

A. Loss Function

In most similar generation models, there is an unavoidable problem, whether MLP architecture or deep convolution network architecture is adopted in GAN. The better the discriminator is trained, the more obvious the gradients vanish of the generator when it iterates. The reason for this is that there are some problems with the definition of the loss function of GAN. We detail these problems in this section.

For any sample, whether from the real sample set or the generated sample set, its contribution to the loss function is as follows in (1):

$$-P_r(x)[\log D(x)] - P_z(z)[\log(1 - D(G(x)))]. \tag{6}$$

By taking the derivative with respect to $D(x)$, it is easy to determine that the optimal discriminator is expressed as follows:

$$D_{optimal}(x) = \frac{P_r(x)}{P_r(x) + P_g(x)}. \tag{7}$$

When the condition of the optimal discriminator is reached, the loss function of GAN is transformed into the following form

$$\begin{aligned}\min_G \max_D V(D,G) &= E_{x\sim P_r(x)}\left[\log \frac{2P_r(x)}{P_r(x)+P_g(x)}\right] \\ &\quad + E_{x\sim P_z(z)}\left[\log \frac{2P_g(x)}{P_r(x)+P_g(x)}\right] - 2log2 \\ &= 2JS(P_r\|P_g) - 2log2. \end{aligned} \tag{8}$$

Thus, when the precision of the discriminator reaches an optimal situation, the minimization of loss function of GAN is equivalent to the minimization of (8). We can then easily determine that the requirement to generate real samples is also transformed into minimizing the Jensen–Shannon divergence between $P_r(x)$ and $P_g(x)$.

However, if $P_r(x)$ and $P_g(x)$ barely overlap or do not overlap at all, the Jensen–Shannon divergence between $P_r(x)$ and $P_g(x)$ is $log2$. When we use the gradient descent to solve the minimum value for a constant, it is not surprising to see the phenomenon of the gradient vanish. Unfortunately, the situation described above highly tends to occur often.

Therefore, by drawing lessons from previous experience, the loss function of our method adopts the Wasserstein distance instead of the Jensen–Shannon divergence. This article tends to use the addition of label information to directionally control the class of generated SAR images. In general, it seems reasonable to adopt the loss function in the following form:

$$\begin{aligned} L \approx \max_{\omega:\|D_\omega\|_L \le K} & E_{x\sim P_r(x)}[D_\omega(x|y)] \\ & - E_{x\sim P_g(x)}[D_\omega(G(x|y))] \end{aligned} \tag{9}$$

where y represents the label information, which is added artificially. However, as exists in all WGAN methods, the networks defined by the above loss function may face the gradient explode problem during the process of training. In order to avoid gradient explode in the training process of the proposed method, a more robust loss function is proposed in this article to solve the above problems in this article. The loss function adopted in this article is expressed as follows:

$$\begin{aligned} L = & E_{x_g\sim P_g}[D(x_g|y)] - E_{x_r\sim P_r}[D(x_r|y)] \\ & + \lambda E_{x\sim P_x}[(\|\nabla_x D(x|y)\|_2 - 1)^2] \end{aligned} \tag{10}$$

where y represents the label information. It is the key to being able to generate SAR images directionally.

B. Architecture of LDGAN

The LDGAN includes a deconvolution network G and a convolution network D. The networks also include the addition of generated image label information. The overview of the architecture of the LDGAN is shown in Fig. 2.

The contents of each model components are introduced below. Table I gives a detailed specification of the LDGAN.

The generator, called G, is a de-CNN. The inputs run over the network and get an expanding output rather than obtaining a low dimensional representation of higher spatial dimension, such as the input of the convolutional networks. Additionally, it should be noted that the filter stride must be smaller than the filter shape. This is to ensure that the output regions of the two filters overlap to form the generated images. At the same time, although the deconvolution network in this article still has a convolutional architecture, it is better to use strided convolution to replace spatial pooling functions. z indicates the random noise, which is the initialization of the LDGAN. y is the real label information of the generated image. It is added to the initialization information in the form of one hot encode to generate the image.

The discriminator, called D, is a CNN. It is similar to the recent CNN that implements the vision tasks. However, its role is not a probability value used to output the decision classification, but to determine whether the image $G(z)$ generated by the generator is a real image. Moreover, to determine whether the generated image is a directed generation, the label information y is also input as a condition to the D in order to determine whether the generated image is a real picture of a certain category. As shown in Fig. 2, the location of insertion of label information is extraordinarily essential. It is not only important information used to generate images, but also establishes an essential foundation for decision boundaries in discrimination.

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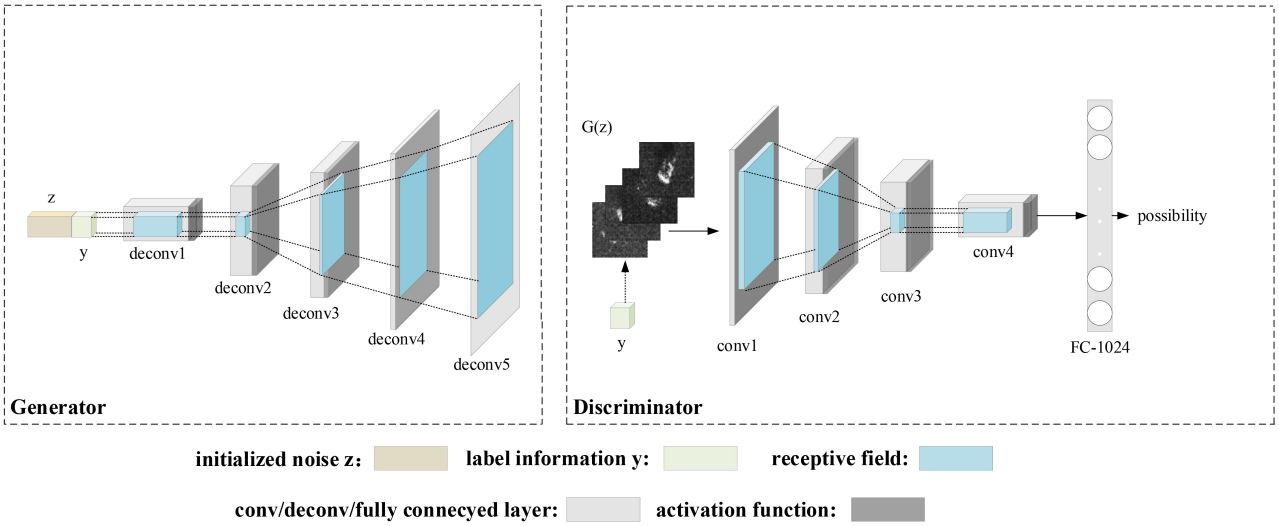


Fig. 2. Overview of the architecture of LDGAN. LDGAN consists of a deconvolution network and a convolution network.

TABLE I
DETAILED SPECIFICATION OF LDGAN

The architecture of label directed GAN					
the generator			the discriminator		
Type	Filter Size/stride	Output Size	Type	Filter Size/stride	Output Size
deconv1	$4 \times 4/1$	$4 \times 4 \times 1024$	conv1	$4 \times 4/2$	$32 \times 32 \times 128$
deconv2	$4 \times 4/2$	$8 \times 8 \times 512$	conv2	$4 \times 4/2$	$16 \times 16 \times 256$
deconv3	$4 \times 4/2$	$16 \times 16 \times 256$	conv3	$4 \times 4/2$	$8 \times 8 \times 512$
deconv4	$4 \times 4/2$	$32 \times 32 \times 128$	conv4	$4 \times 4/2$	$4 \times 4 \times 1024$
deconv5	$4 \times 4/2$	$64 \times 64 \times 1$	FC-1024	-	1

G consists of five deconvolution layers. In the G, every filter size is 4×4. However, there is no fully connected, or pooling layer are used in these de-CNNs. Instead, the strided deconvolutions, often referred to as fractional-strided convolutions in some articles, are proposed to replace the pooling layers. Most of the filter stride values are 2, however, the filter stride is 1 in the deconvolution layer 1. This is to ensure the size of the generated image and the ability to generate images at the generator.

The D consists of four convolution layers and one fully connection layer. Obviously, in the last layer, the full connection layer is proposed instead of using the sigmoidal function. The purpose of the original GAN discriminator is to implement a binary classification for a real and fake sample. Thus, it is reasonable to use the sigmoidal function in the last layer. However, the purpose of the discriminator of the LDGAN is changed to approximate the Wasserstein distance between the real distribution and the generated distribution. The goal becomes a regression task, so it is necessary to remove the sigmoidal function. Besides, there are four convolution layers for feature extraction before the full connection layer. All the value of filter strides is 2. The filter size of all the filters are 4×4, and the pooling layer after the convolution layer is replaced by the strided convolutions, as it is beneficial to allow the network learn to its spatial down sampling.

C. Training Details

Training the GAN with high performance standards often requires delicate tricks. In the process of training, there are many hyperparameters that need to be adjusted. Even different optimization algorithms make a significant difference in GAN performance. Each type of GAN often requires different training techniques in order to ensure high-quality performance.

For the original GAN, the mini-batch stochastic gradient descent was first used to train GAN. Besides, the number of steps to apply to the discriminator per generator iteration is set to 1, which is the most convenient option. However, the original GAN is based on the MLP architecture. It is difficult to dig deeper image features.

With wide application of the convolution network with the image, it is natural to scale up GAN using CNN to generate the images. However, the training method of DCGAN is different from the training method of GAN. First, the networks change the activation function to LeakyReLU and set the slope of all LeakyReLU functions to 0.2 in the network. Next, this method specifies the parameters of weights initialization and the size of the mini-batch. Subsequently, the DCGAN utilizes the Adam algorithm to optimize the model, and it also makes strict rules on the learning rate and momentum term β_1.

Owing to the intractable problems of DCGAN, the emergence of WGAN also brings more training tricks to the GAN. Consequently, changing the size of the mini-batch and the value of the learning rate also increases the number of iterations of the discriminator when the generator iterates one time. In addition, when the parameters of the discriminator are updated per iteration, the parameters need to be truncated to a specific range, which makes the network prone to gradient explode. Of course, the obvious difference between this kind of networks and other types of GAN lies in the selection of optimization algorithms. The WGAN utilizes the root mean square prop (RMSprop) algorithm rather than the Adam algorithm for the model optimization.

According to prior training experience of all types of the GAN and the architecture of the LDGAN, the detailed algorithm of the LDGAN is shown in Algorithm 1.

Here, we will provide a more detailed explanation of the parameter set of the proposed algorithm in this article. The setting of the parameter set can be divided into the following

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Algorithm 1 LDGAN, Our Proposed Algorithm. All the Experiments in This Article Use Default Values of $\lambda = 10$, $n_{diter} = 1$, $\alpha = 0.0002$, $\beta_1 = 0.5$, $\beta_2 = 0.9$, $m_{batch} = 100$,$D_{att} = 0.97$, $G_{att} = 0.99$

Require: The coefficient for GP λ, the number of steps to apply to the discriminator per generator iteration n_{diter}, the learning rate α, the hyperparameters of the Adam algorithm β_1,β_2, the batch size m_{batch}, the attenuation rate of learning rate of discriminator D_{att}, the attenuation rate of learning rate of generator G_{att}

Require: The initial generator parameters ω_g, the initial discriminator parameters ω_d. All the initial parameters must be on the premise that the mean is 0 and the variance is 0.02.

Require: The one-hot coding based on the sample label y

while ω_g has not converged **do**
  for i $= 1, \ldots, n_{diter}$ **do**
    for j $= 1, \ldots, m_{batch}$ **do**
      Sample the real data $x_r \sim P_r(x)$, the prior sample $z \sim P_z(z)$
      Sample a random number $\xi \sim U[0, 1]$
      Combine the one-hot coding y with the input of D and G
      $x_g \leftarrow G(z|y)$
      $x \leftarrow \xi x_r + (1 - \xi)x_g$
      $L_i \leftarrow D(x_g|y) - D(x_r|y) + \lambda(\|\partial \frac{D(x|y)}{\partial x}\|_2 - 1)^2)$
    end for
    $\omega_d \leftarrow Adam(\frac{1}{m}\sum_{i=1}^{m} \partial \frac{L_i}{\partial \omega_d}, \omega_d, \alpha, \beta_1, \beta_2, D_{att})$
  end for
  Sample a batch of prior samples $(z^i)_{i=1}^{m_{batch}} \sim P_z(z)$
  Combine the one-hot coding y with the input of D and G
  $\omega_g \leftarrow Adam(-\frac{1}{m}\sum_{i=1}^{m} \partial \frac{D(G(z|y)|y)}{\partial \omega_g}, \omega_g, \alpha, \beta_1, \beta_2, G_{att})$
end while

three parts. First, the parameters related to generation result of the proposed algorithm include four parameters: λ, α, β_1, and β_2. These parameters are all validated in other articles. According to the improved WGAN [24], the experiments use $\lambda = 10$, which the authors found to work well across a variety of architectures and data sets ranging from toy tasks to large ImageNet CNNs. Moreover, the GP imposed on the WGAN is necessary. Because it is easy to lead to ignoring the higher moments in the data distribution when the WGAN [23] adopts weight clipping to avoid the phenomenon of mode collapse. Weight clipping means that each time the parameters of the discriminator have been updated, their absolute values are truncated to not more than a fixed constant c. It is noteworthy that the lower moment is related to the overall characteristics of the image and does not contain much detailed information of the image. At the same time, our network architecture fuses the characteristics of the improved WGAN and DCGAN [27]. Therefore, it is necessary to draw on DCGANs successful experience in parameter setting. The suggested learning rate of 0.001 is too high for the training process of DCGAN, thus the learning rate is set to 0.0002. Furthermore, the authors found setting the momentum term β_1 to 0.5 helps stabilize the training process and avoid oscillations. The setting of the momentum term β_2 is the default selection of the Adam algorithm parameters. The parameters related to the convergence speed of algorithm include three parameters m_{batch}, D_{att}, G_{att}, and m_{batch}, which are flexible parameters selected according to the graphics processing unit (GPU) memory, and the value of the parameters can be freely set according to the hardware configuration of the running platform. This affects only the speed of the algorithm and does not affect subsequent generation results and recognition results. D_{att} and G_{att} are two parameters about the attenuation of the learning rate. This operation is the same as the purpose of other learning rate attenuation, which ensures that the model does not fluctuate too much in the later stages of training process, thus bringing the results closer to the optimal solution. Moreover, the consideration of attenuation of the learning rate is also essential. As SAR image generation progresses gradually, the convergence of loss function becomes more and more difficult. The lower learning rate can guarantee the convergence of loss function. Finally, there are different options for parameter setting of n_{diter} in different articles. Therefore, we also perform different generation verification experiments for the two parameter settings mentioned in many articles. According to our experimental results, the number of steps for applying the discriminator per generator iteration is important as it has a direct impact on the generation results. The size of this parameter affects the precision of the discriminator, which indirectly guides the change of the result of generator of the LDGAN. The generation results do not benefit from too many iterations over the discriminator of the LDGAN, so the value of n_{diter} is set to 1.

IV. Experimental Results

The experiments are carried out from the two aspects of authenticity and reliability of the generated image and the practical significance of the generated image in this article. The authenticity and reliability of image generation reflects the ability of the LDGAN to dig through and learn image

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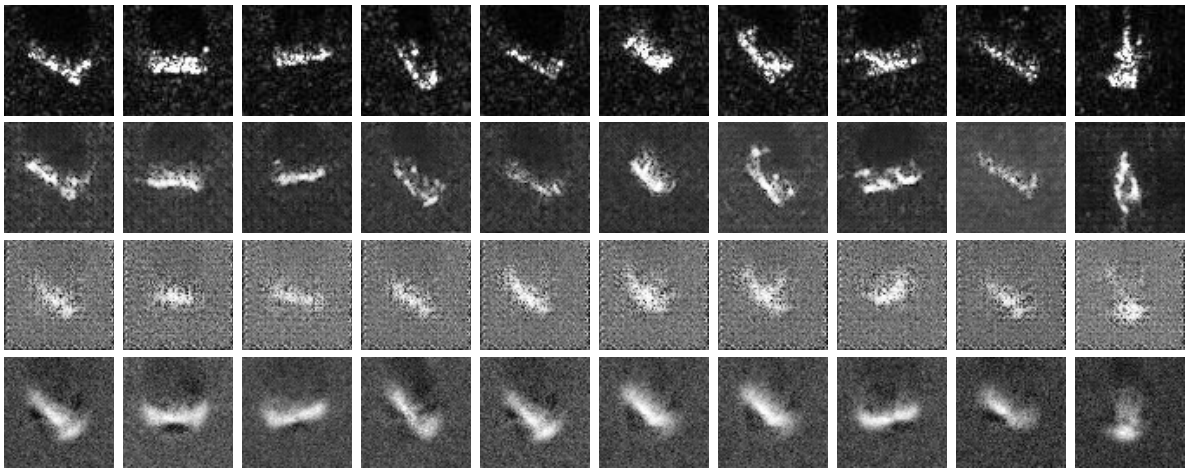


Fig. 3. Comparison of generation results of ten categories SAR target samples. The order from left to right of each row of images is as follows: 2S1, BMP2, BRDM2, BTR60, BTR70, D7, T62, T72, ZIL131, and ZSU23/4. Row 1: real images of ten categories SAR target samples. Row 2: images generated by the LDGAN for ten categories SAR target samples. Row 3: images generated by the conditional deep convolutional GAN for ten categories SAR target samples. Row 4: images generated by conditional GAN for ten categories SAR target samples.

information. Authenticity refers to the difference perceived from different viewpoints between the generated image and the real image. Reliability is the reasonable degree that the generated images are used for target recognition. However, it is not enough to generate a target image with high authenticity and reliability in the field of target recognition. We also aim to explore the potential application of the generated image to actual demand. Thus, it is necessary to implement experiments researching the practical significance of an image generated by the LDGAN.

In order to prove the effectiveness of the proposed method, all experimental results were dependent on the data of the Moving and Stationary Target Acquisition and Recognition (MSTAR) database in this article. Ten categories of SAR target samples were selected in the process of these experiments. The samples used in this experimental process were slices of MSTAR from ten SAR categories; all the slices were 64×64 RAW data. The training samples are the target samples with a pitch angle of 17°. In addition, the experiment chose the target samples with a pitch angle of 15° as test samples.

A. Authenticity and Reliability of Generated Images

In the field of optics, due to the high interpretability of optical images, the authenticity of the generated optical images often are directly perceived by the naked eye. However, SAR target images are much less interpretable than optical images. It is not enough to rely solely on experience to intuitively judge the authenticity of the generated image. Therefore, in addition to the display results of generated images, this experiment also demonstrates the authenticity of generated images from the statistical features and gradient features of images.

At the same time, to objectively reflect the comparison between the proposed method and other methods, the experiment also added two different methods that can directionally generate images as a comparison to experimental results. Thus, in addition to comparing the generated image by the LDGAN with the real image, this experiment also took the generation results of condition GAN and condition DCGAN together as a comparison of the experimental results.

Gray histograms can represent the number of pixels with a particular shade of gray in an image and reflect the frequency of the occurrence of an image reflected by the particular shade. For SAR images, the distribution of gray histograms also reflects the statistical characteristics of SAR images.

The comparison of generation results between the three directional generation methods and the real image is shown in Fig. 3. Fig. 3 demonstrates that conditional GAN only learns some contour features due to its MLP architecture. In contrast, it is obvious that the GAN with deep convolutional architecture digs out more detailed features of SAR images. However, the loss functions of the conditional GAN and the conditional deep convolutional GAN do not utilize the Wasserstein distance so that it is easy for the collapse mode to occur in the process of the training.

The phenomenon of collapse mode is shown in Fig. 4. The images generated by the conditional deep convolutional GAN and conditional GAN are only a single type of sample. The generated sample lacks diversity, which is the obvious phenomenon of collapse mode. As shown in Fig. 4, through the comparison of results from two different types of images, several things can be observed. First, the proposed method effectively avoids the collapse mode problem. The other two generation methods are prone to attain the collapse mode during the training process. Second, from the visual and intuitive experience, the images generated by the LDGAN are most similar with the real images. However, due to the poor interpretability of SAR images, a comparison of experimental results uses the statistical characteristics of three categories of generated images and the real images.

The related experimental results are shown in Fig. 5. This experiment shows the results of five categories of images including 2S1, BMP2, BRDM2, BTR60, and BTR70. As shown in Fig. 5, the distribution of the gray histogram of

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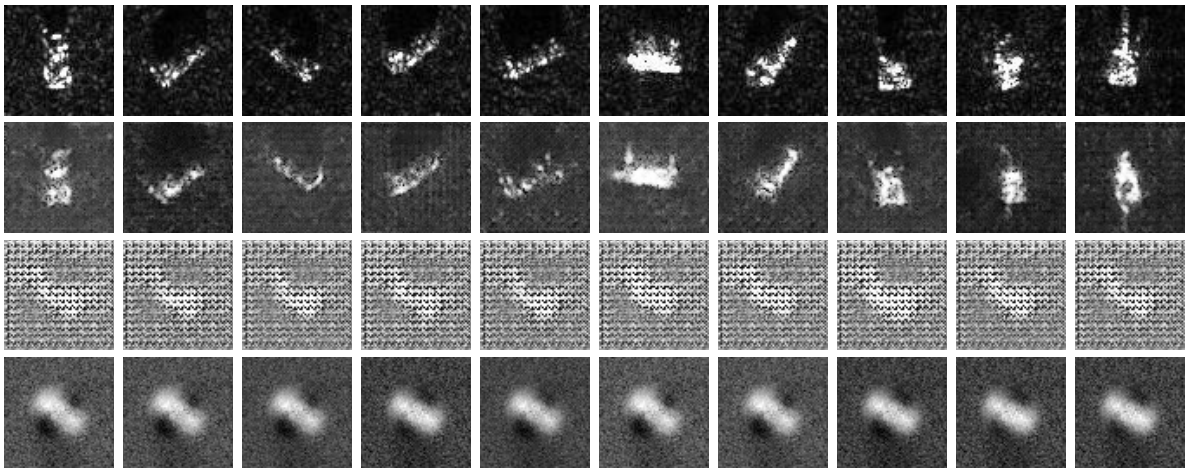


Fig. 4. Comparison results of collapse mode of ten categories SAR target samples. The order from left to right of each row of images is as follows: 2S1, BMP2, BRDM2, BTR60, BTR70, D7, T62, T72, ZIL131, and ZSU23/4. Row 1: real images of ten categories SAR target samples. Row 2: images generated by the LDGAN for ten categories SAR target samples. This type of method effectively avoids the problem of collapse mode during the training process. Row 3: results of the image generated by such a method are displayed in row 3 when the conditional deep convolutional GAN encounters the collapse mode problem. Row 4: results of the image generated by such a method are displayed in row 4 when the conditional GAN encounters the collapse mode problem.

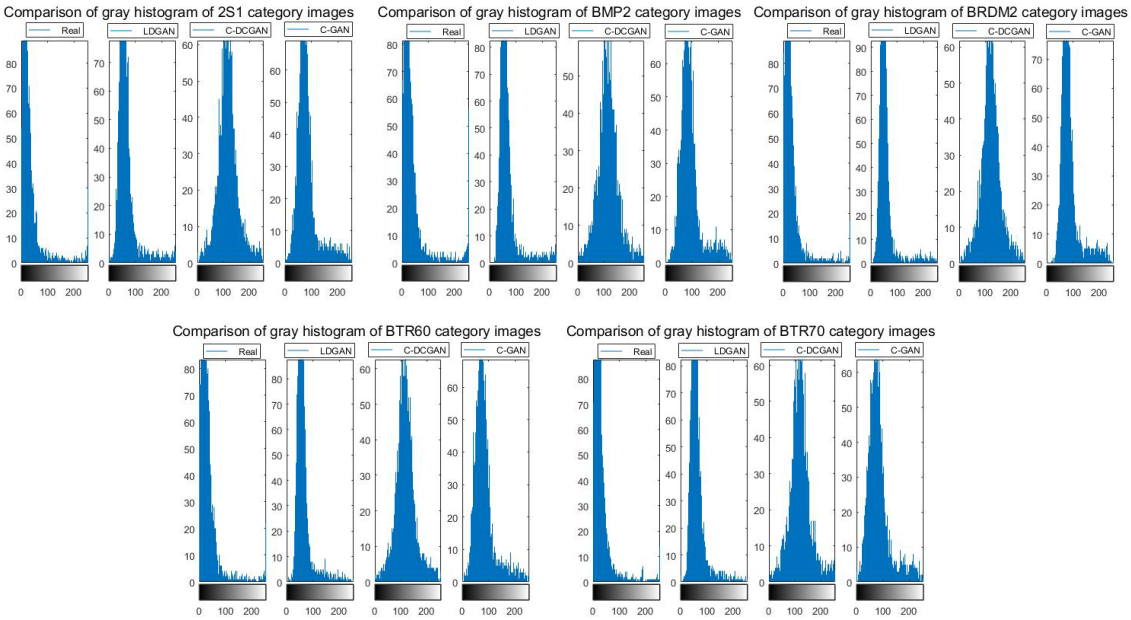


Fig. 5. Gray histogram comparison results of four types of images. In each graph, the histogram of grayscale statistics of images from four different sources in a specific category of images is drawn. The order from left to right of each graph is as follows: the gray histogram of the real image, the gray histogram of the image generated by the LDGAN, the gray histogram of the image generated by the conditional DCGAN, and the gray histogram of the image generated by the conditional GAN. All target categories included in the figure are as follows: 2S1, BMP2, BRDM2, BTR60, and BTR70.

the images generated by the conditional DCGAN is notably different from the histogram of the images from other sources. In terms of the shape of the gray histogram, the proposed method is also closer to the real image than the image generated by conditional GAN. However, compared with the original image, the gray value of the image generated by the LDGAN increased overall, which is shown as right shift in each graph. This also demonstrates from the statistical characteristics that the image generated by the LDGAN is roughly the same in shape as the original image, but the overall image looks brighter. It is not enough to compare the statistical characteristics between images from four different sources by direct observation. In addition, in order to make the results of this experiment more convincing, we also adopted the histogram similarity index to measure the distance between different histograms in each graph. Table II shows the results of similarity comparison between gray histograms of images from four different sources. It is not difficult to find that the gray histogram of the images generated by the LDGAN is most similar to that of the real image. By observing the comparative results of gray histogram and comparing the values of histogram similarity index, it is not difficult to prove that the image generated by the LDGAN is the closest to the real image in statistical characteristics.

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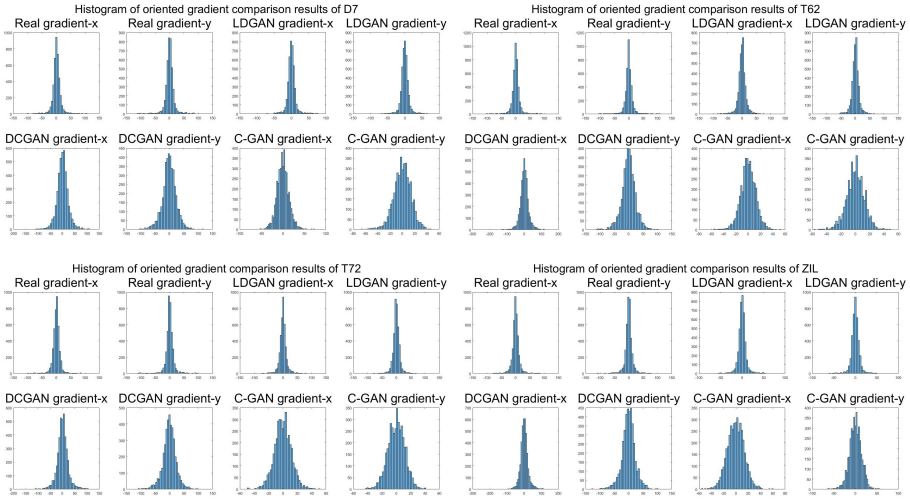


Fig. 6. Histogram of oriented gradient comparison results of four types of images. In each graph, the histogram of gradient features of images from four different sources in a specific category of images is drawn. Row 1: order from left to right of each graph is as follows: the gradient along the x-direction of the real image, the gradient along the y-direction of the real image, the gradient along the x-direction of the image generated by the LDGAN, the gradient along the y-direction of the image generated by the LDGAN. Row 2: order from left to right of each graph is as follows: the gradient along the x-direction of the image generated by the conditional DCGAN, the gradient along the y-direction of the image generated by the conditional DCGAN, the gradient along the x-direction of the image generated by the conditional GAN, the gradient along the y-direction of the image generated by the conditional GAN. All target categories included in the figure are as follows: D7, T62, T72, and ZIL131.

TABLE II

HISTOGRAM SIMILARITY INDEXES OF THE THREE TYPES OF GENERATED IMAGES AND REAL IMAGES

Index: Comparative object / The type	LD&real	DC&real	Conditional&real
$2S1$	0.3643	0.2119	0.2934
$BMP2$	0.3545	0.2417	0.2638
$BRDM2$	0.3898	0.1813	0.1997
$BTR60$	0.3518	0.2015	0.3210
$BTR70$	0.3415	0.1827	0.3185

However, it is not enough to measure the statistical characteristics of the image as a whole. In our experiments, we also considered it important to focus on whether the appearance and shape of the local target are learned by the generation model. In an image, the appearance and shape of the local target are usually well described by gradients. Thus, the gradient features of images from four different sources were needed for comparison.

The oriented gradient histogram compares results from four types of images shown in Fig. 6. Different from the above experiment, this experiment shows the results from the other four categories of images including D7, T62, T72, and ZIL131. According to our experimental results, the images generated by the conditional DCGAN and the images generated by the conditional GAN is worse than the images generated by the LDGAN about the purpose of learning the local appearance and shape of the MSTAR target. In other words, the gradient information of the MSTAR target image generated by the LDGAN is better than the two other methods whether it is the gradient along the x-direction or y-direction. Unfortunately, the oriented gradient histogram of the images generated by the LDGAN shows that the images of this type still fail to fully learn the gradient information of the real image. This observation is particularly obvious for the categories of the T62 and ZIL. These results also imply that the image produced by the generation method shown in this article cannot currently replace the real image.

Similarly, in order to more rigorously prove the availability of the proposed method, this experiment also numerically compares the histograms of oriented gradient in each graph in Fig. 6. However, the gradient histogram similarity comparison in this experiment are compared separately along the x-direction and the y-direction. Table III shows the results of similarity comparison between histograms of gradient of images from four different sources.

From the numerical results, the histogram of oriented gradient of the image generated by the LDGAN is most similar to that of the real image. In other words, according to the direct observation results of Fig. 6 and the quantitative comparison results of Table III, the image generated by the LDGAN is most similar to that of the real image on the gradient information.

Besides, one of the issues discussed at the beginning of this article is that most machine learning algorithms are the data-driven algorithms. As an MSTAR target image generated by GAN, the next set of experiments also explore the potential of such images as training samples for the target recognition.

However, before we determined whether the appearance and shape of the local target are learned by the generation model, we first conducted an experiment which would make a preliminary judgment on the reliability of the generated samples. Most machine learning algorithms are based on historical data, and the models trained by the historical data can fit future data well. Thus, it is necessary to use historical data containing the overall representation. By establishing the assumption that samples are independent and identically

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TABLE III
RESULTS OF SIMILARITY COMPARISON BETWEEN HISTOGRAMS OF GRADIENT OF IMAGES FROM FOUR DIFFERENT SOURCES

	The comparison results along the x-direction			The comparison results along the y-direction		
Target Type	LD&real	DC&real	Conditional&real	LD&real	DC&real	Conditional&real
D7	0.9991	0.9546	0.9546	0.9840	0.9720	0.9805
T62	0.9736	0.9520	0.9662	0.9871	0.9637	0.9834
T72	0.9983	0.9565	0.9689	0.9936	0.9649	0.9808
ZIL	0.9897	0.9564	0.9613	0.9892	0.9723	0.9725

TABLE IV
MI VALUES OF THE THREE TYPES OF GENERATED IMAGES AND REAL IMAGES

MI: Comparative object / The type	LD&real	DC&real	Conditional&real
2S1	1.70	1.92	1.81
BMP2	1.66	2.06	1.94
BRDM2	1.32	1.80	1.59
BTR60	1.47	1.88	1.81
BTR70	1.47	1.84	1.84
D7	1.51	1.98	1.84
T62	1.61	1.93	1.65
T72	1.50	1.82	1.50
ZIL131	1.25	1.75	1.60
ZSU23/4	1.49	1.77	1.63

distributed, we can reduce the situation of individual cases in the training sample. The generation models based on the GAN are all designed to fit the generated sample distribution and the real sample distribution, however, the distance measurement is different across different methods. Therefore, to use the generated samples for target recognition, it is sufficient to evaluate the independence between the generated samples and the real samples.

In the information theory, mutual information (MI) of two random variables is a measure of interdependence between variables. The degree of dependence between two samples can also be represented by the value of MI. This experiment selects 40 images of four types as shown in Fig. 3. Because the generated samples are based on a data set of real samples, this experiment evaluates the MI values between the three types of generated images and the corresponding real images. This kind of experimental processing also provides reliable proof for the degree of dependence between each type of the generated samples and the real samples.

The comparison results of the MI values of the three types of generated images and real images are shown in Table IV.

In Table IV, the slash table header consists of three parts: type, comparative object, and MI. Type refers to the type of the target, which is ordered as follows: 2S1, BMP2, BRDM2, BTR60, BTR70, D7, T62, T72, ZIL131, and ZSU23/4. The comparative object indicates the comparisons of three types of generated images and real images, respectively. MI shows the MI values of different comparative objects in different targets.

According to Table IV, the performance of MI value of the target image generated by the LDGAN and the real image is significantly better than the performance of the MI values of the other two types of generated images and the real image. Because the MI value of the two images is smaller, the degree of interdependence between these images is smaller. In other words, the independence between the two images is more evident. Therefore, the image generated by the LDGAN is more independent of the real image than the other two types of generated images.

At the same time, to further explore the reasonable degree of the generated images for ATR, the experiment also introduced a set of classical features for the SAR target for comparison. These classical features include scale and shape features, contrast features, and texture features. Although electromagnetic features are an important component of the classical features of the SAR target, when using GAN as a model for generation, the method is generally aimed at fitting the distribution of the SAR target image. SAR imaging is an inversion method from echo to target scattering information. The image generation method proposed in this article is a method from image to image, and there is a significant difference between the SAR imaging and the proposed algorithm. In other words, the complex image of the SAR target is not obtained during the image generation process, so this experiment does not consider electromagnetic features as one of the objects for comparison in the experimental results.

The experiment divides the SAR target from the target slice. In this experiment, the images shown in Fig. 3 are selected as the data of this experiment. Four sets of segmented targets are selected for a subsequent feature comparison. The order of the segmented targets is as follows: 2S1, BTR70, T62, and ZIL131. As shown by the results of target segmentation in Fig. 7, when the generated target is segmented from the original image, the target generated by the LDGAN is still most similar to the real target. The comparative results of classical features for the SAR target based on the segmented targets are shown in Tables V and VI.

From the numerical results of Tables V and VI, the classical features of SAR target recognition of images generated by the LDGAN is the closest than that of the real images. In other words, the segmented targets generated by the LDGAN are still most similar to the real targets in scale and shape, contrast, and texture features. Thus, we also confirmed the reliability of using images generated by the LDGAN for ATR using the comparison of a series of classic features of SAR targets.

In summary, we demonstrated the authenticity of the images generated by the LDGAN through a series of experiments. These experiments also detected that the image generated

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TABLE V
COMPARISON RESULTS OF AVAILABLE FEATURES FOR 2S1 AND BTR70

target category		2S1				BTR70			
Feature Type	Typical Characteristics	Real	LDGAN	Conditional DCGAN	Conditional GAN	Real	LDGAN	Conditional DCGAN	Conditional GAN
scale and shape features	mass	355	328	246	574	372	372	317	486
	Moment of Inertia	10.77	10.66	7.16	11.78	11.23	11.23	10.50	10.16
Contrast Features	Peak of CFAR	127	127	127	127	127	127	127	127
	Average of CFAR	123.49	123.59	127	126.62	107.82	114.11	127	126.99
Texture Features	Standard Deviation	4.83	5.14	4.06	4.42	4.31	4.31	6.38	4.05
	Fractal Dimension	-0.95	-0.73	-0.98	-1.29	-1.11	-1.11	-0.078	-1.35

TABLE VI
COMPARISON RESULTS OF AVAILABLE FEATURES FOR T62 AND ZIL131

target category		T62				ZIL131			
Feature Type	Typical Characteristics	Real	LDGAN	Conditional DCGAN	Conditional GAN	Real	LDGAN	Conditional DCGAN	Conditional GAN
scale and shape features	mass	428	427	326	534	335	385	213	370
	Moment of Inertia	10.99	10.76	9.77	11.09	11.12	11.24	7.02	9.95
Contrast Features	Peak of CFAR	127	127	127	127	127	127	127	127
	Average of CFAR	125.41	125.52	127	126.99	103.42	122.70	127	122.70
Texture Features	Standard Deviation	4.62	3.76	6.35	3.79	4.88	4.69	6.05	3.88
	Fractal Dimension	-1.16	-1.26	-0.30	-1.36	-1.01	-1.21	-0.30	-1.29

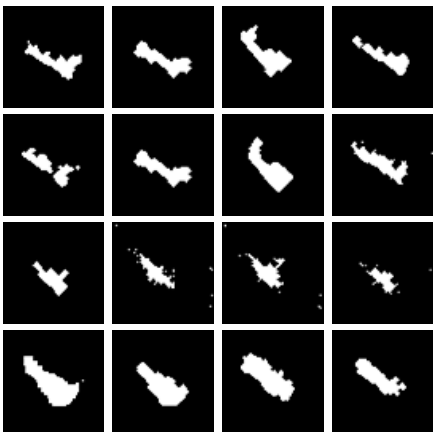


Fig. 7. Comparison results of target segmentation of four categories SAR target samples. The order from left to right of each row of images is as follows: 2S1, BTR70, T62, and ZIL131. Row 1: target segmentation of the real images. Row 2: target segmentation of the images generated by the LDGAN. Row 3: target segmentation of the images generated by the conditional deep convolutional GAN. Row 4: target segmentation of the images generated by the conditional GAN.

by the LDGAN is more in line with the assumptions of independent and identically distributed samples than the other two types of generated images. In addition, the experiment proved that the proposed method performs better on a series of classical features of the generated SAR targets. The images generated by the LDGAN show excellent experimental results regarding authenticity and reliability. Thus, we subsequently explored the potential application of the generated image to actual demand using our proposed method.

B. Performance of Generated Images for ATR

From the previous set of experiments, it is apparent that the three types of generated pictures learn the relevant information of the target image to a certain extent. However, the experiments demonstrate lack of clarity on whether the images generated by the learned information have application potential in ATR. Therefore, we conducted a set of verification experiments to investigate on this issue.

This set of experiments is carried out separately using two distinct parts. The first part of the experiment explores the usability of the generated image in target recognition. The second part of the experiment provides clarifying data, which allow for discussion on the potential of the application of the generated image to the actual problem of the SAR target recognition.

In the first set of experiments, the general idea is to add generated samples to the training samples, which will then reflect changes in the model recognition rate, allowing us to confirm the role of the generated samples by retraining the recognition model. However, if the information of the training sample enables the recognition model to learn thoroughly, the changes in the recognition rate becomes more subtle, which makes it difficult to identify that these changes of the experiment are caused by the generated samples. This can also be caused by other factors, such as whether the generated sample is added to the training sample or if initialization is different. Thus, it is worth noting that in order to fully reflect the value of the information carried in the generated image, and we must base the premise of the recognition

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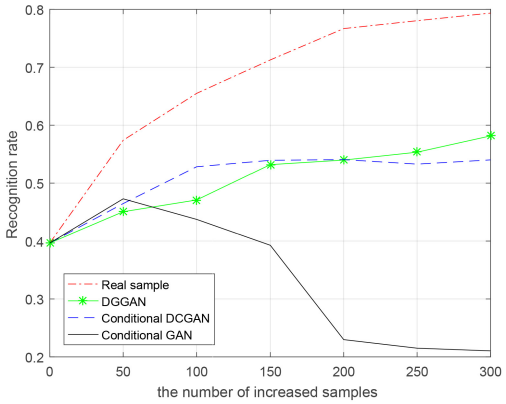


Fig. 8. Comparison results of the recognition rate for the four types of images. All the training samples are used to train the generation models.

experiment on the condition that the recognition model is seriously underfitting.

In the first experiment of this set of experiments, the first step is to select randomly 100 samples from real training samples. Next, we use 50 samples as a batch, and add real samples and three types of generated samples to the training samples in batches to retrain the model. The purpose of the experiment is to observe how the different types of training samples eventually affect the recognition model. Experimental results can be found in Fig. 8.

As visualized in Fig. 8, three types of generated images play a role in supplementing the information to some extent when the model information is insufficient. However, when more than 100 images generated by the conditional GAN are added to the training sample, the recognition rate noticeably decreases. When compared with the real images, the other two types of generated images also have significant gaps in the ability to supplement information within the model.

Therefore, based on the results of this experiment, we began to consider the reason for why the image generated by the conditional GAN has this particular influence on the recognition rate. We next constructed an experiment using the recognition rate of the recognition model obtained from the image generated by the condition GAN as a training sample.

The experiment naturally considers the rationale of using such generated samples for target recognition. However, it is difficult for the naked eye to determine whether all generated images are reasonable with high confidence because of the characteristic of interpretability of SAR images. Therefore, this experiment pretrained an SVM to determine the correctness of all such generated samples. This experiment uses a pretrained SVM to recognize generated samples in order to observe the experimental results. Unfortunately, most of the target images generated by the condition GAN cannot be correctly recognized using the results from the SVM.

Therefore, the recognition rate trend caused by the samples generated by the conditional GAN as a training sample is explained by the above experiment. The image generated by the conditional GAN has available information for recognition. When the sample information is scarce, the increase in the information that can be used for recognition is significant for the recognition rate. With the gradual increase in the number of samples, the lack of sample information is alleviated. At the same time, more useless or even wrong recognition information is added to the training samples. Thus, when such generated samples are added to the training samples, the recognition rate increases significantly when the sample size is small and decreases as the sample size gradually increases.

Naturally, this experiment also uses the above SVM to recognize the other two types of generated images. Our experiments showed that these two types of generated images also experience the phenomenon that some samples cannot be recognized. It should be noted that the probability the image generated by the LDGAN can be correctly recognized by the model is still the highest. More interestingly, the number of the remaining two types of generated images that can be correctly recognized by the recognition model is almost the same. However, the convolutional network extracts the detailed features of the target better due to the CNN characteristics. Thus, we can say that the recognition rate of the model performs better when using the image generated by the conditional DCGAN as the training sample.

However, one problem that cannot be ignored is that the generated image cannot be the same as the real image. Therefore, no matter how strong the ability to extract image information or how likely the generated image can be used for recognition, it is impossible to avoid this problem for a generation model. Moreover, since some of the generated images cannot be correctly recognized, it is also possible for the LDGAN to generate redundant or even incorrect information into the training samples.

Therefore, we attempted to find a more robust way to utilize generated image for target recognition. In the experiment we conducted, the generated image is first filtered by the pretrained SVM, and then the image selected by the filter model is used for target recognition. In order to minimize the impact of intermediate steps on experimental results, all pretrained SVMs adopt the classical SVM architecture. There are not any special tricks for the pretrained SVM. The experimental results are shown in Fig. 9. The image data of the other two types of generated images that can be correctly recognized are very limited, and it is difficult to get a large enough data set to be used as the comparison. As shown in Fig. 9, the generated image filtered by the SVM is used as a training sample, and the trained recognition model does have better performance.

So far, we know that the generated images can complement information, and we have shown how to apply the generated images for better target recognition. However, the above experiments are all based on training samples. When the training samples already have enough recognition information, it is meaningless to generate auxiliary information. Therefore, in the second set of experiments, we also explored whether the generated samples can supplement the information when the number of training samples is insufficient.

To address this question, we randomly selected 400 training samples from the training samples set for generating models. The purpose of the experiment was to observe whether the images produced by the generated model have the ability

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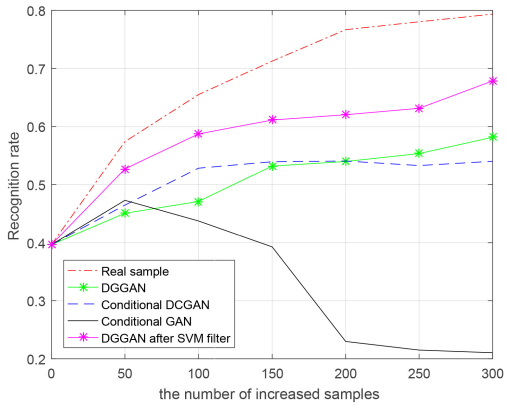


Fig. 9. Comparison results of the recognition rate for the five types of images. All the training samples are used to train the generation models.

TABLE VII
IMPROVEMENT OF THE RECOGNITION RATE OF THE RECOGNITION MODEL TRAINED BY THE GENERATED IMAGES THROUGH LDGAN

number of increased samples	100	200	300	400	500	600
recognition rate	84.68%	84.81%	86.21%	86.05%	85.47%	86.05%
increased recognition rate	5.32%	5.45%	6.85%	6.69%	6.11%	6.69%

to supplement information when using only 400 training samples.

In this experiment, we first implemented data enhancement by flipping and rotating the training samples in order to avoid overfitting for the proposed method. At the same time, the generated image was filtered by the SVM, and then the filtered image was used for the target recognition. The LDGAN used the Wasserstein distance to measure the distance between the generated sample distribution and the real sample distribution. The other two generation methods previously mentioned consistently have significant limitations since they use the Jensen–Shannon divergence as the distance measurement. More importantly, in practical applications, we rarely choose a generation method that may fall into the collapse mode during the process of training. Thus, we only show the experimental results of the proposed method in this experiment.

In Table VII, the experimental results show an improvement of the recognition rate of the model by generating the target image through the LDGAN using only 400 training samples. According to prior experience, the highest recognition rate the trained SVM model can achieve using the 400 training samples on this recognition task is about 79.36%. However, after using the proposed method to supplement information, the recognition rate significantly improved. The experimental results prove that the proposed method still can supplement information when the sample information is insufficient. However, it is worth noting that the method proposed in this article is not aimed at solving the problem of the insufficient number of samples in the target recognition task but should still be addressed in future studies. In our present study, our focus was on the generation ability of the proposed generation model as well as researching prospects for the improvement of potential of generated samples.

V. CONCLUSION

Data-driven recognition methods play an increasingly important role in current methods for target recognition. Inspired by the recent success of GAN in generating samples, this article proposes a novel generation model, called LDGAN, to alleviate the problem that training sample information is insufficient for target recognition. This article gives a series of explanations on the derivation, architecture, and characteristics of the LDGAN. First, the detailed derivation of the loss function of the LDGAN is presented in this article. Second, we provided some explanations of the adjustments in the corresponding architecture regarding some theoretical issues in the derivation process. Finally, we outlined a detailed algorithm for the specific implementation of the LDGAN to generate images. We expound on a series of experiments, which also verified the authenticity and reliability of images generated by the LDGAN. We have proven experimentally that the image generated by the LDGAN does complement the useful information for the target recognition. More importantly, we also obtain a way to use the generated images for target recognition more effectively through our experiments. Through comparative experiments, we also demonstrated that the LDGAN can complement the sample information when the training sample information is insufficient. Experimental results show the various advantages of the proposed generation model and the necessity of the proposed method to directionally generate images. The proposed method also provides some potential for solving the problem of insufficient training sample information in ATR.

REFERENCES

[1] Y. Sun, Z. Liu, S. Todorovic, and J. Li, "Adaptive boosting for SAR automatic target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 43, no. 1, pp. 112–125, Jan. 2007.

[2] Z. Jianxiong, S. Zhiguang, C. Xiao, and F. Qiang, "Automatic target recognition of SAR images based on global scattering center model," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3713–3729, Oct. 2011.

[3] J. A. O'Sullivan, M. D. DeVore, V. Kedia, and M. I. Miller, "SAR ATR performance using a conditionally Gaussian model," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 1, pp. 91–108, Jan. 2001.

[4] U. Srinivas, V. Monga, and R. G. Raj, "SAR automatic target recognition using discriminative graphical models," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 50, no. 1, pp. 591–606, Jan. 2014.

[5] C. Lardeux *et al.*, "Support vector machine for multifrequency SAR polarimetric data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 12, pp. 4143–4152, Dec. 2009.

[6] Q. Zhao and J. C. Principe, "Support vector machines for SAR automatic target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 2, pp. 643–654, Apr. 2001.

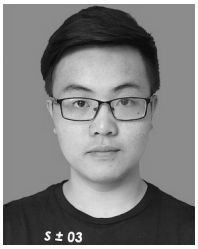
[7] V. V. Chamundeeswari, D. Singh, and K. Singh, "An analysis of texture measures in PCA-based unsupervised classification of SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 2, pp. 214–218, Apr. 2009.

[8] A. Lima, H. Zen, Y. Nankaku, C. Miyajima, K. Tokuda, and T. Kitamura, "On the use of kernel PCA for feature extraction in speech recognition," *IEICE Trans. Inf. Syst.*, vol. 87, no. 12, pp. 2802–2811, 2004.

[9] J. Yang, D. Zhang, A. F. Frangi, and J.-Y. Yang, "Two-dimensional PCA: A new approach to appearance-based face representation and recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 26, no. 1, pp. 131–137, Jan. 2004.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:01:53 UTC from IEEE Xplore. Restrictions apply.

[10] Z. Cui, Z. Cao, J. Yang, J. Feng, and H. Ren, "Target recognition in synthetic aperture radar images via non-negative matrix factorisation," *IET Radar, Sonar Navigat.*, vol. 9, no. 9, pp. 1376–1385, Dec. 2015.
[11] T. Cooke, "Two variations on Fisher's linear discriminant for pattern recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 2, pp. 268–273, Feb. 2002.
[12] S. Pang, S. Ozawa, and N. Kasabov, "Incremental linear discriminant analysis for classification of data streams," *IEEE Trans. Syst., Man, Cybern. B, Cybern.*, vol. 35, no. 5, pp. 905–914, Oct. 2005.
[13] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.
[14] S. A. Wagner, "SAR ATR by a combination of convolutional neural network and support vector machines," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 52, no. 6, pp. 2861–2872, Dec. 2016.
[15] Y. Zhou, H. Wang, F. Xu, and Y.-Q. Jin, "Polarimetric SAR image classification using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1935–1939, Dec. 2016.
[16] X. Liu, Y. Huang, J. Pei, and J. Yang, "Sample discriminant analysis for SAR ATR," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 12, pp. 2120–2124, Dec. 2014.
[17] D. Malmgren-Hansen, A. Kusk, J. Dall, A. A. Nielsen, R. Engholm, and H. Skriver, "Improving SAR automatic target recognition models with transfer learning from simulated data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1484–1488, Sep. 2017.
[18] B. Ding, G. Wen, X. Huang, C. Ma, and X. Yang, "Data augmentation by multilevel reconstruction using attributed scattering center for SAR target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 6, pp. 979–983, Jun. 2017.
[19] J. Guo, B. Lei, C. Ding, and Y. Zhang, "Synthetic aperture radar image synthesis by using generative adversarial nets," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 7, pp. 1111–1115, Jul. 2017.
[20] N. Merkle, S. Auer, R. Müller, and P. Reinartz, "Exploring the potential of conditional adversarial networks for optical and SAR image matching," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 6, pp. 1811–1820, Jun. 2018.
[21] F. Gao, Y. Yang, J. Wang, J. Sun, E. Yang, and H. Zhou, "A deep convolutional generative adversarial networks (DCGANs)-based semi-supervised method for object recognition in synthetic aperture radar (SAR) images," *Remote Sens.*, vol. 10, no. 6, p. 846, 2018.
[22] C. P. Schwegmann, W. Kleynhans, B. P. Salmon, L. W. Mdakane, and R. G. V. Meyer, "Synthetic aperture radar ship discrimination, generation and latent variable extraction using information maximizing generative adversarial networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 2263–2266.
[23] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, Jul. 2017, pp. 214–223.
[24] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. C. Courville, "Improved training of wasserstein gans," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5767–5777.
[25] M. Zhang, Z. Cui, X. Wang, and Z. Cao, "Data augmentation method of SAR image dataset," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 5292–5295.
[26] I. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.
[27] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," Nov. 2015, *arXiv:1511.06434*. [Online]. Available: https://arxiv.org/abs/1511.06434
[28] M. Mirza, and S. Osindero, "Conditional generative adversarial nets," Nov. 2014, *arXiv:1411.1784*. [Online]. Available: https://arxiv.org/abs/1411.1784
[29] G. Antipov, M. Baccouche, and J. L. Dugelay, "Face aging with conditional generative adversarial networks," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 2089–2093.
[30] M. Arjovsky and L. Bottou, "Towards principled methods for training generative adversarial networks," Jan. 2017, *arXiv:1701.04862*. [Online]. Available: https://arxiv.org/abs/1701.04862
[31] X. Chen, Y. Duan, R. Houthooft, J. Schulman, I. Sutskever, and P. Abbeel, "InfoGAN: Interpretable representation learning by information maximizing generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 2172–2180.
[32] L. Tran, X. Yin, and X. Liu, "Disentangled representation learning GAN for pose-invariant face recognition," in *Proc. CVPR*, Jul. 2017, vol. 3, no. 6, pp. 1283–1292.



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